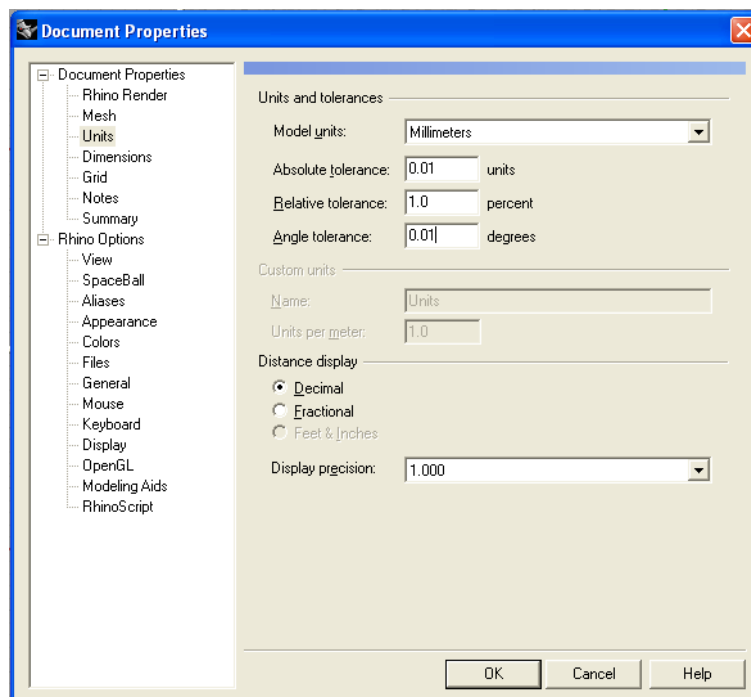


HowTo Rhino & ICEM

Simple 2D model

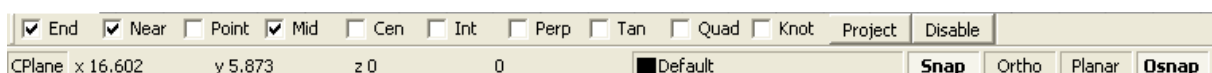
- 1) New file setup: choose Millimeter (automatically converts to Meters if imported to ICEM)
- 2) Set units: File → Properties → Units:
 Model units: should already be Millimeters
 Abs. tolerance = size of smallest feature (treated as Meters by CSMP) = i.e. 0.01
 Rel. tolerance = ratio between smallest and largest object (in %) = i.e. 1.0
 Angle tolerance = similar to abs. tolerance, but for angles = i.e. 0.01 (SKM: 0.1)
Question: does it have to be Millimeters? → No, because CSMP treats it as Meters anyway!



- 3) Draw around origin (best centered!), otherwise significant digits can be lost (Y = up, X = left-right → draw in “TOP” view for 2D)

Doesn't matter if box  or line  drawing (must be NURBS curves → check in Object Properties!)

Use snapping to exactly close lines at intersections:



- 4) Assign objects to certain layers (to e.g. separate fractures from bedding): select several objects with SHIFT → right-click on layer → “Select Object Layer”
- 5) Split a curve: select the curve → command “Split” or  → select cutting objects (do this with all intersections!), use cut curves to cut the other ones (“cutting the other way round”)

Explode closed curves with 
 (Tool in ICEM: “Topology” → splits automatically, but might destroy things one still needs!)

- 6) ICEM needs to know where the beginning and end of objects are (→ puts node there): create start and end points for all curves: select all curves → Curve → Point Object → Mark Curve Start & End, create points at intersections: Curve → Curve From Objects → Intersection (or command “Intersection”)

Question: are multiple points a problem? To what layer do they have to belong to?

(To remove duplicates: Edit → Select Objects → Duplicate Objects → DELETE)

- 7) Rename layers: LEFT, RIGHT, TOP, BOTTOM (doesn't have to be upper case, ICEM automatically converts into upper case) → put boundary lines into specific layers (IRREGULAR not possible in 2D!!!)

- 8) Create one surface that covers all objects: Select all → Surface → Planar Curve

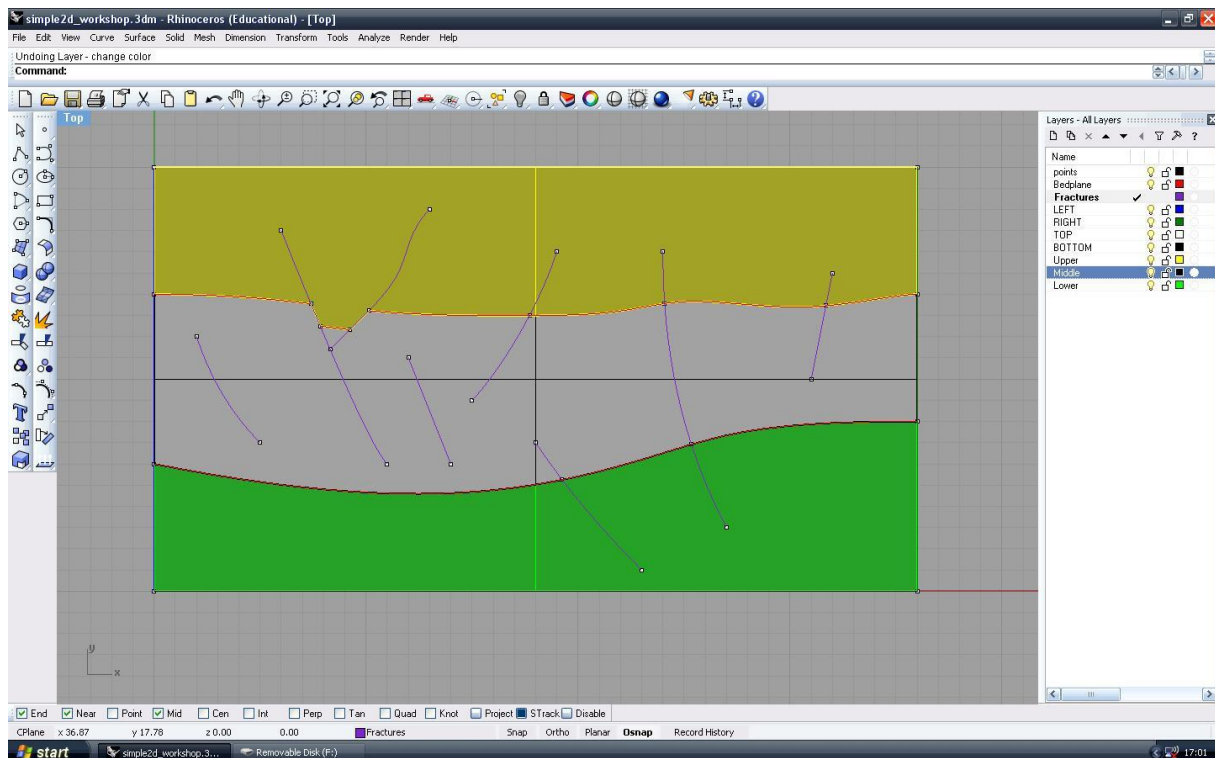
- 9) Split the surface: select the surface → command “Split” or  → select the curves that should cut it

Move new surfaces to separate (new) layers

(Splitting of surfaces by fractures that do not cut all through is not possible)

Question: do we need to draw outlining curves to the surfaces?

(To show surfaces in layer colors: Right-click on View corner “TOP” → Shaded)



- 10) Save as Rhino version 3 (.3dm) – watch out: all layers of objects that should be exported, should be visible (light bulb on)!

- 11) Drag&drop the file onto 3dm2tin.exe (ANSYS 12 makes it automatically itself!)

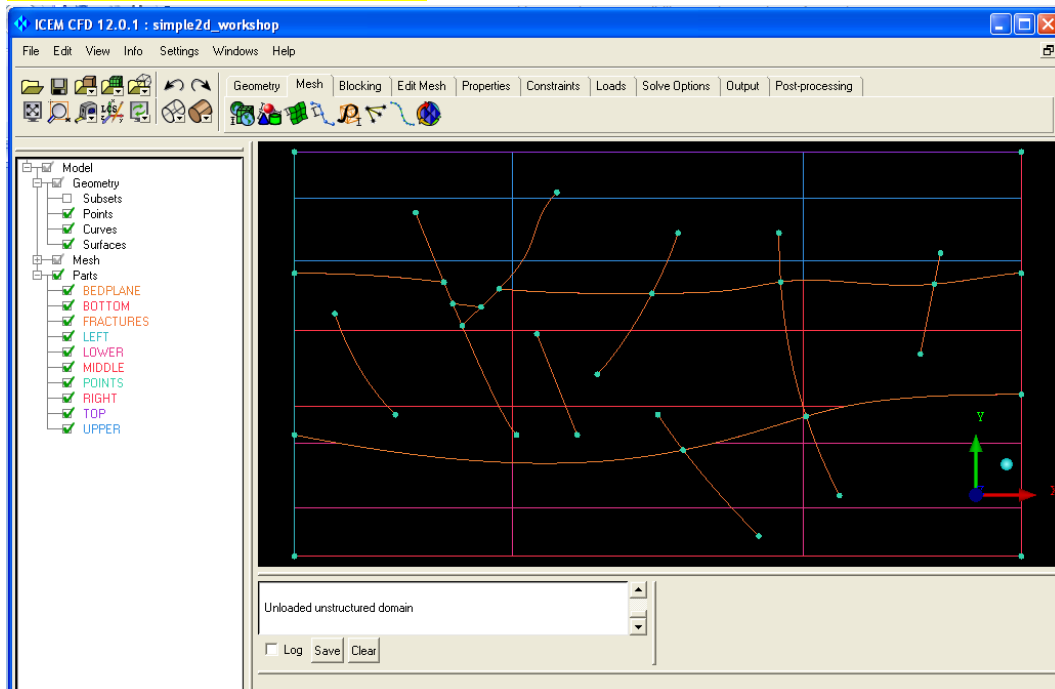
ICEM (screenshots from ANSYS 12)

- 1) Open the geometry in ICEM CFD:

ANSYS 11: File → Geometry → Open Geometry (choose .tin)

ANSYS 12 (?): File → Import Geometry → Rhino 3DM (choose .3dm; ICEM automatically creates a .tin) → check Use Object Names for Parts and Include Hidden Layers → click Apply

Question: File from Rhino v3 or v4?



(left window: "Parts" contain objects from the Rhino layers, with according labels)

- 2) Change the appearance of objects, if necessary:

Right-click on "Geometry" objects

- 3) If a line/surface/... is missing (e.g. bottom line), draw your own line/surface/...:

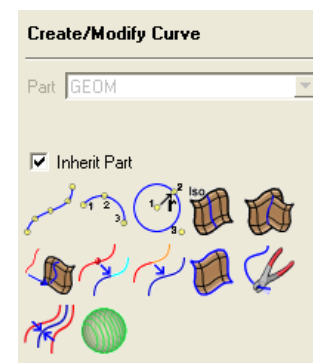
Right-click on "Parts" → Create Part → call it BOTTOM

Select the two corner points → draw a line with one of the

Curve tools:



(sometimes do actions several times (because of bugs!!!!))



- 4) Moving objects into another layer/"part":

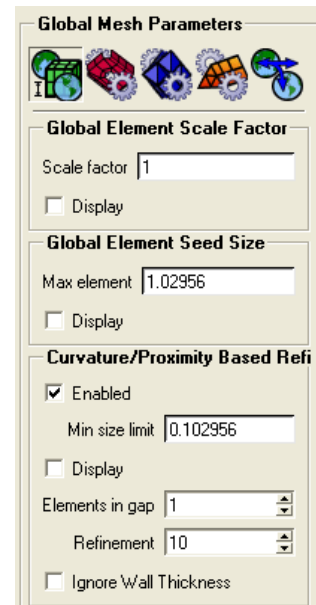
Right-click on part → Add to Part → click on arrow next to the Entity field (in section Add to Part by Selection) → select line type for entity → select the line with left-click → accept with middle-click (or undo with right-click)

5) Global Mesh Setup (change default values, if necessary):



(#1)

1. Global: Max element: if small, mesh will be fine – if large, mesh will be coarse (= max. size a mesh element will have)
2. Surface: Mesh type = e.g. All Tri (depending on what kind of mesh elements is wanted), Mesh Method = Patch Independent (= most robust)
3. Volume (only for 3D): Mesh type = Tetra/Mixed, Mesh Method = Robust (Octree)
4. Grow prism layers (only for 3D, or if uniform mesh needed) (check “Display” to show e.g. resulting min/max element sizes)



6) Mesh sizes for parts:



(#2) → table

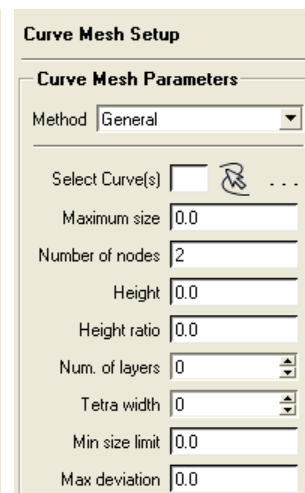
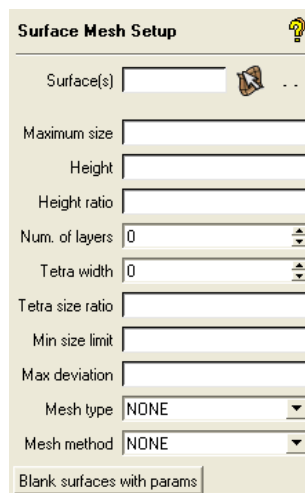
Check fractures as “Int Wall”, because they are objects with one dimension less, that need to have other properties assigned

7) Surface and Curve Mesh Setup:



(#3 and 4)

For surfaces: Select a surface → accept with middle-click → Maximum size = e.g. 0.4 (= max. element size in meters) → click Apply (Similar rules apply for curves)

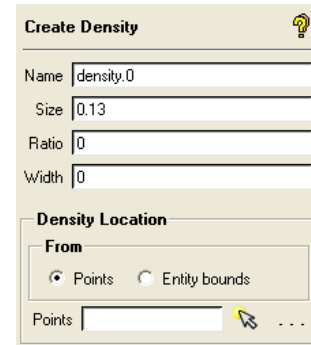


- 8) To refine certain parts of the mesh, create a "Density":



(#5)

Either draw with points or do it via entity boundaries



- 9) Compute Mesh:



(#8)

Possible to overwrite global settings → click Compute

(for 3D: option to run it with batch process in Global Volume Meshing Parameters; step 5)

- 10) Visualize as solid, turn off all geometries → detect errors visually while looking at it from different directions

- 11) Check mesh for quality with Edit Mesh:



(#3 and 4)

Check important options (in 3D also check Error: Volume/Surface orientations; look up in Wiki)

→ click Apply → single-edge: if at boundary, it is OK → click Ignore

(Quality graph show aspect ratios: small aspect ratio means bad quality)

- 12) Smooth the mesh globally (#5 in Edit Mesh) has already been done automatically → graph shows quality (x axis) vs. number of elements (y axis) → selecting column of bad elements highlights bad element in the mesh

- 13) Repair mesh, e.g. if there is a hole:



(#7)

- 14) If another mesh should be computed:

File → Mesh → Close mesh, then create a new one

15) Assign boundary conditions and output the mesh:



(#1, 2 and 4)

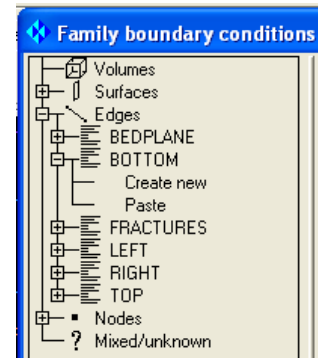
Output (#1) → select “CSP” output solver

(+NASTRAN structural solver)

Boundary condition (#2) → select the boundary (e.g. BOTTOM)

→ Create new → select boundary node id → pick corresponding node flag (e.g. BOTTOM OUTSIDE; it's possible to do this just for one, then it recognized all the others) → click Copy and Accept

Write input (#4) → choose your project → write name of output file (as BINARY!) → click Done → creates .asc / .dat / .log



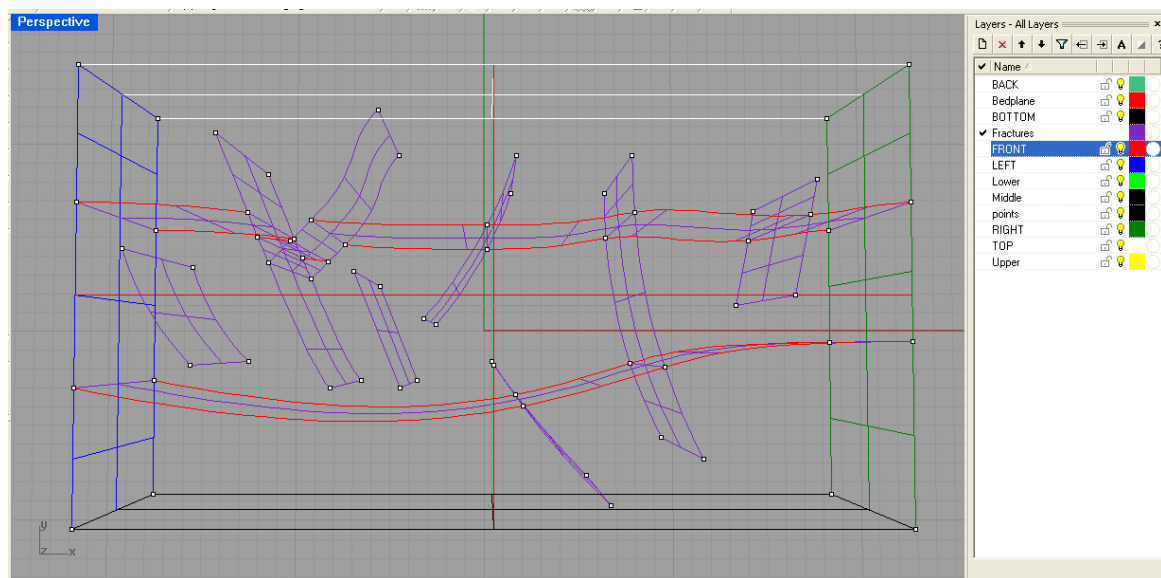
16) .asc and .dat are required for CSMP

(Look up the Help-file for meshing options!)

Simple 3D model (by extruding the simple 2D model in Rhino)

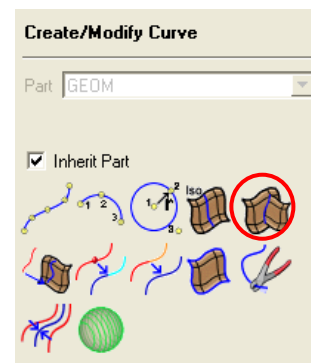
Rhino:

- 1) Select all curves: Edit → select objects → all curves
- 2) Extrude: Surface → extrude curve → straight → type in number or click with mouse
- 3) Copy all lines, points and surfaces from one reference point (corner in front) to other reference point (corner in back) – copies are put into the currently active layer, therefore check that!!!
- 4) Create two new layers (FRONT and BACK) and move the corresponding surfaces there
- 5) Move the whole block, so that it is positioned as centered on the origin
- 6) Generate intersection curves in ICEM



ICEM:

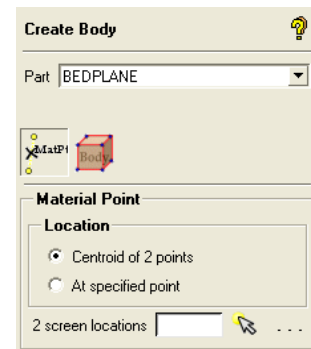
- 1) Open 3D geometry in ICEM
- 2) Check intersections (e.g. between bedding plane and fracture, bedding planes and model boundaries, etc.):
Surface to surface intersection → select all planes → middle-click → click Apply → new part "GEOM" is created, containing the intersections)



- 3) Define bodies by setting a point into the volumes so that CSP can distinguish between different volumes and assign different properties:



Create Body (Geometry, #4) → type in a new name for a new part (e.g. middlebed) → select button “Material Point” → e.g. Centroid of 2 points → select two points that are diagonally positioned within the volume (if there are no diagonally positioned points, create a new point on one of the volume edges) → middle-click to accept → click Apply
(A single volumes that is entirely cut e.g. by a fracture will have to be defined as two separate bodies → give them the same name!)



- 4) Define fractures as internal walls

Question: Do intersection have to be defined as internal walls?

- 5) Check whether all objects are in the appropriate part, because mesh properties will be assigned to specific parts; otherwise move (e.g. bedding plane surfaces from part “FRACTURES” to part “BEDPLANE”, because in Rhino they were automatically assigned to the same layer as the fractures!):

Right-click on part where to move to → Add to Part → select all surfaces → middle-click → click Accept

- 6) Compute mesh (as for 2D)

- 7) Check the mesh & mesh quality

→ ignore messages about multiple edges (at intersections?) and single edges (at boundaries), delete unconnected vertices

Question: What is the reason for multiple/single edges? What are important checking options that should not be missed?

(for better visibility: right-click on part (e.g. mesh shells) → Solid & Wire)

- 8) Output mesh (as for 2D)